

CHURAPCHA, Russian Federation

WMO: 247680

Lat: 62.033N

Lon: 132.600E

Elev: 186

StdP: 99.11

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-52.1	-50.5	-54.0	0.0	-51.8	-52.3	0.0	-50.2	4.0	-31.4	3.4	-32.6	0.0	220	0.697

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	30.3	18.5	28.3	17.7	26.2	17.1	20.1	27.0	19.1	25.5	18.0	24.2	2.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	13.1	22.2	16.6	12.1	22.0	15.4	11.2	20.9	58.6	26.9	55.3	25.6	51.6	24.2	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.2	5.3	4.4	-53.4	32.7	2.2	1.9	-55.0	34.1	-56.3	35.3	-57.5	36.3	-59.2	37.7		
(5)			-53.4	22.3	2.2	2.2	-55.0	23.9	-56.3	25.2	-57.5	26.4	-59.2	28.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.6	-40.3	-35.3	-20.9	-5.0	7.3	15.8	18.8	14.9	5.6	-8.7	-28.7	-40.2	(6)
(7)		DBStd	22.46	6.11	6.19	7.59	6.26	4.26	4.13	3.84	3.90	4.42	7.11	9.11	6.43	(7)
(8)		HDD10.0	7793	1561	1267	958	451	104	5	1	6	144	578	1161	1557	(8)
(9)		HDD18.3	10283	1819	1501	1216	701	341	96	42	121	383	837	1411	1816	(9)
(10)		CDD10.0	642	0	0	0	0	22	179	273	157	11	0	0	0	(10)
(11)		CDD18.3	89	0	0	0	0	0	20	56	13	0	0	0	0	(11)
(12)		CDH23.3	1182	0	0	0	0	6	327	689	159	1	0	0	0	(12)
(13)		CDH26.7	361	0	0	0	0	0	86	239	36	0	0	0	0	(13)
(14)	Wind	WSAvg	1.6	0.7	0.7	1.4	2.0	2.6	2.3	2.1	1.9	1.9	1.7	1.1	0.7	(14)
(15)	Precipitation	PrecAvg	258	6	5	5	8	23	38	53	44	30	24	15	7	(15)
(16)		PrecMax	396	19	15	16	22	55	88	120	89	65	51	37	18	(16)
(17)		PrecMin	176	1	1	1	0	1	8	12	4	2	5	3	1	(17)
(18)		PrecStd	51	4	4	4	6	13	22	27	22	15	10	8	4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-18.7	-12.4	2.8	12.2	23.9	31.4	33.8	30.4	21.6	6.8	-2.1	-18.6	(19)
(20)			MCWB	-19.1	-12.9	-0.6	5.2	13.0	18.7	19.9	19.0	13.2	4.0	-3.1	-19.0	(20)
(21)		2%	DB	-24.9	-18.2	-1.4	9.1	20.9	29.1	31.2	26.9	17.8	3.6	-6.4	-24.8	(21)
(22)			MCWB	-25.2	-18.6	-4.0	3.2	11.1	17.4	19.1	18.0	11.4	1.1	-7.1	-25.0	(22)
(23)		5%	DB	-28.2	-22.3	-4.6	6.7	18.5	27.1	29.4	24.9	15.2	1.7	-11.3	-27.7	(23)
(24)			MCWB	-28.4	-22.6	-6.6	1.9	9.8	16.3	18.3	16.8	10.0	-0.1	-11.9	-27.9	(24)
(25)		10%	DB	-31.4	-25.6	-7.7	4.4	16.0	25.0	27.4	22.8	13.0	-0.1	-15.1	-31.0	(25)
(26)			MCWB	-31.6	-25.9	-9.2	0.4	8.5	15.4	17.8	15.7	9.3	-1.3	-15.5	-31.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-19.1	-12.8	-0.7	5.5	13.8	20.1	22.0	20.7	14.0	4.2	-2.9	-19.0	(27)
(28)			MCDB	-18.7	-12.2	2.7	11.5	22.2	28.0	29.2	25.9	19.5	6.3	-2.2	-18.6	(28)
(29)		2%	WB	-25.1	-18.7	-3.9	3.6	11.9	18.5	20.6	19.2	12.5	1.4	-7.1	-25.0	(29)
(30)			MCDB	-24.9	-18.4	-1.5	8.7	19.6	26.3	28.1	24.9	16.2	3.0	-6.5	-24.8	(30)
(31)		5%	WB	-28.3	-22.6	-6.6	2.1	10.2	17.4	19.7	17.9	11.2	0.0	-11.9	-27.9	(31)
(32)			MCDB	-28.2	-22.3	-4.7	6.2	17.2	25.0	26.5	23.4	14.2	1.3	-11.3	-27.7	(32)
(33)		10%	WB	-31.5	-25.9	-9.2	0.7	9.0	16.3	18.8	16.5	9.6	-1.3	-15.5	-31.1	(33)
(34)			MCDB	-31.4	-25.6	-7.7	4.0	15.0	23.4	25.1	21.5	12.5	0.0	-15.1	-31.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.0	12.0	17.1	13.6	11.0	12.2	11.7	11.3	9.6	8.0	8.1	5.9	(35)
(36)			MCDBR	9.9	13.7	17.2	13.0	15.6	16.2	15.7	14.7	14.0	8.6	10.1	9.4	(36)
(37)			MCWBR	9.7	13.3	14.8	8.4	7.7	6.5	5.7	6.6	8.1	6.3	9.8	9.4	(37)
(38)		5% WB	MCDBR	10.0	13.7	16.7	12.1	14.0	14.8	12.9	12.3	11.2	7.6	9.9	9.6	(38)
(39)			MCWBR	9.9	13.2	14.5	8.1	7.4	6.7	5.6	6.1	6.9	5.9	9.6	9.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.207	0.255	0.299	0.359	0.442	0.406	0.488	0.435	0.346	0.284	0.215	0.178	(40)	
(41)		taud	1.979	2.081	2.066	2.030	2.042	2.290	2.030	2.160	2.382	2.285	1.989	1.889	(41)	
(42)		Ebn at Noon	480	702	796	813	769	818	724	734	755	670	448	315	(42)	
(43)		Edh at Noon	37	71	105	136	147	117	149	120	79	58	35	22	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.37	1.37	3.06	4.92	5.28	5.89	5.46	4.22	2.56	1.18	0.50	0.17	(44)	
(45)		RadStd	0.02	0.06	0.11	0.19	0.45	0.43	0.32	0.31	0.28	0.11	0.04	0.01	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		+0.75	+1.58	+1.20	N/A	N/A	N/A	-234	-262	+81	N/A	
(47) Regional (5 neighbors)		+0.71	+1.43	+1.26	N/A	N/A	N/A	-236	-246	N/A	N/A	

Nomenclature: See separate page